

Contents

1 History of Macroeconomics

1.1 Phase 0: Panics of the 1800s and early 1900s

1.2 Phase 1: Measuring Macroeconomic Activity (1930s to Early 1950s)

During the Great Depression, Maynard Keynes published the General Theory of Employment, Interest, and Money in 1936. The basic tenet of the “Keynesian view” was that various “rigidities” plague market transactions and lead to potentially long-lasting disequilibrium outcomes. The clearest way to understand Keynesianism was that nominal wages and/or prices may not adjust quickly enough to clear quantity supplied and quantity demanded. Hence Keynesian logic demands that the government should and is able to aid the economy.

In order to do so, there needed to be some US-wide measures of the performance of markets; until the Great Depression, there were none. The system of GDP accounting that we more or less still continue to use began during the Great Depression. The concepts and measurements of aggregate GDP and aggregate consumption and aggregate investment that we now take for granted in the typical basic macroeconomics-class GDP accounting equation were essentially invented during the Depression.

By the mid-1940s, economists had collected and tabulated about 15 years of quarterly data (roughly 60 time periods) on what now are considered “standard macro measures” often used to judge that society’s standard of living. There are two basic findings. First, in the long-time horizon there is a steady upward march of GDP (long-term trend). Second, there are many ups and downs along this long-run path, called business cycle.

Should there be one unified framework to consider both long-run growth and business cycle fluctuation of the economy? The answer to this question is no. Understanding, both empirically and theoretically, the long-run growth component is often referred to as the branch of economic growth or economic development; understanding the short-run fluctuations of the economy is almost universally referred to as the branch of macroeconomics.

1.3 Phase 2: Keynesian Macroeconometric Approach (Early 1950s to late-1970s)

The end of WWII is often attributed to the powerful engineering and technology the US military rapidly developed. This is also the period in which newly developed mainframe computers using punch cards to process computations were being increasingly put to research use.

The mathematically and computationally based ideas crept into macroeconomic thinking. Not uncoinciden-

tally, much of the high-powered technology that aided military efforts was developed at research universities at which top-notch economists were also housed.

Starting in the early 1950s, a heavy dose of mathematics and statistical analytics using high-powered mainframes quickly became the fashion of macroeconomics, first in academic circles and then, by the early 1960s, in policy circles. A paramount goal for these emerging computable statistical description of aggregate economic events—the Keynesian macroeconomic approach—was to answer an important question hanging over macroeconomics: How can business cycle be “explained”?

Throughout the 1950s many Keynesian macroeconomic framework were developed around the world. One prominent large-scale Keynesian macroeconomic frameworks that quickly took center stage was the MIT/Penn/Federal Reserve Board model constructed by a consortium of researchers and policy advisers at these three institutions.

Championing this effort was a group of three prominent economists from academe, all of whom would be future Nobel laureates—Paul Samuelson (Nobel recipient in 1970), James Tobin (Nobel recipient in 1981), and Robert Solow (Nobel recipient in 1987). Samuelson, Tobin, and Solow were not cloistered academics in the Ivory Tower. Each spent significant time in his career serving in various government positions that advised President John F. Kennedy and President Lyndon B. Johnson, including serving on the White House Council of Economic Advisers.

The Keynesian-inspired macroeconomics models took the form

$$\begin{aligned}x_{1t} &= \alpha_0 x_{2t} + \alpha_1 x_{3t} + \alpha_2 x_{4t} + \cdots \\x_{2t} &= \alpha_3 x_{1t} + \alpha_4 x_{3t} + \alpha_5 x_{4t} + \cdots \\&\cdots \\x_{136t} &= \alpha_{597} x_{1t} + \alpha_{598} x_{13t} + \alpha_{599} x_{69t} + \cdots ,\end{aligned}$$

where α describes the correlation observed in the real world between an economic variable on the RHS and the economic variable on the LHS.

The US economic growth was incredibly strong during the 1950s and 1960s, perhaps in part due to the “great policy tips” provided by the estimated models (from Keynesian approach). It came to the point where macroeconomic ups and downs started to be considered a solved problem.

This seemed to be true throughout the 1950s and 1960s—but it then turned out to be no longer true in the stagflationary period of the 1970s and early 1980s. All of a sudden, the glory decades of macroeconomics of the 1950s and 1960s seemed to have collapsed. If macroeconomics were to remain an organized field

of thought, scores of economists figured that the future of macro had to somehow depart from Keynesian macroeconometrics because there was something seemingly inconsistent with economic analysis in these frameworks.

Many researchers struggled to describe the essence of this inconsistency. Finally, in 1976, it was the economist Robert Lucas (future Nobel recipient in 1995) who simply and elegantly described the root of the issue. His (later named) Lucas critique pointed out that the α coefficients were essentially never seriously thought of as being dependent on policy. A much larger, much deeper issue arose from the Lucas critique, which is that Keynesian macroeconomic models are not economic models, but rather only statistical descriptions of economic outcomes.

There are alternative ways of “defining” economics, but the theme that runs through them all is that economics studies how individuals make informed choices given scarce resources. Do the α terms capture these ideas? The answer was no. The macro profession was in disrepute by the late 1970s, on the verge of extinction.

1.4 Phase 2.5: The Rise of Monetarism

The rise of the monetarist school of thought bridged “phase 2” and “phase 3” of macroeconomic thought. Milton Friedman examined every downturn of the US economy since the mid-1800s and found that they were all accompanied by either an outright decrease in the level of the supply of money or a decrease in the growth rate of the supply of money.

Friedman’s analysis led to the conclusion that the Federal Reserve should not have clamped down on the nominal money supply in the early stages of the Depression. Reagan found Friedman’s descriptions clear. Friedman was to become an economic consultant for Reagan’s campaign for President in 1980.

Reagan won in a landslide against Jimmy Carter’s re-election campaign, and Paul Volcker essentially built directly on the monetarist idea, by clamping down tightly on the issuance of new nominal money to bring down the inflation in the 1980s.

In terms of the proper reach of macroeconomic policy and how prescriptive policy should be, monetarism (sometimes known by the name “quantity theory of money”) ideologically bridged the decline of the Keynesian macroeconometrics phase in the 1970s with the emergence of the microeconomic-based approach of modern macroeconomics of the late 1970s and 1980s.

1.5 Phase 3: Modern Macroeconomic Framework (late 1970s to Present)

Modern macroeconomics begins by explicitly studying the microeconomic principles of utility maximization, profit maximization, and market clearing (micro-foundation). This modern macroeconomic approach quickly captured the attention of the profession through the 1980s for two reasons. First, it actually begins with microeconomic principles. Second, it was in the early 1980s that desktop computing started to become widespread. The new breed of modern macroeconomic frameworks could thus be computed directly in one's office, rather than needing to reserve time for use of costly mainframe machines.

Theories:

- The real business cycle theory (RBC): consumption-leisure model, consumption-savings model, ...
- The New Keynesian theory, which resuscitates Keynes's ideas of "stickiness" in nominal prices, but in modern general-equilibrium macro form.

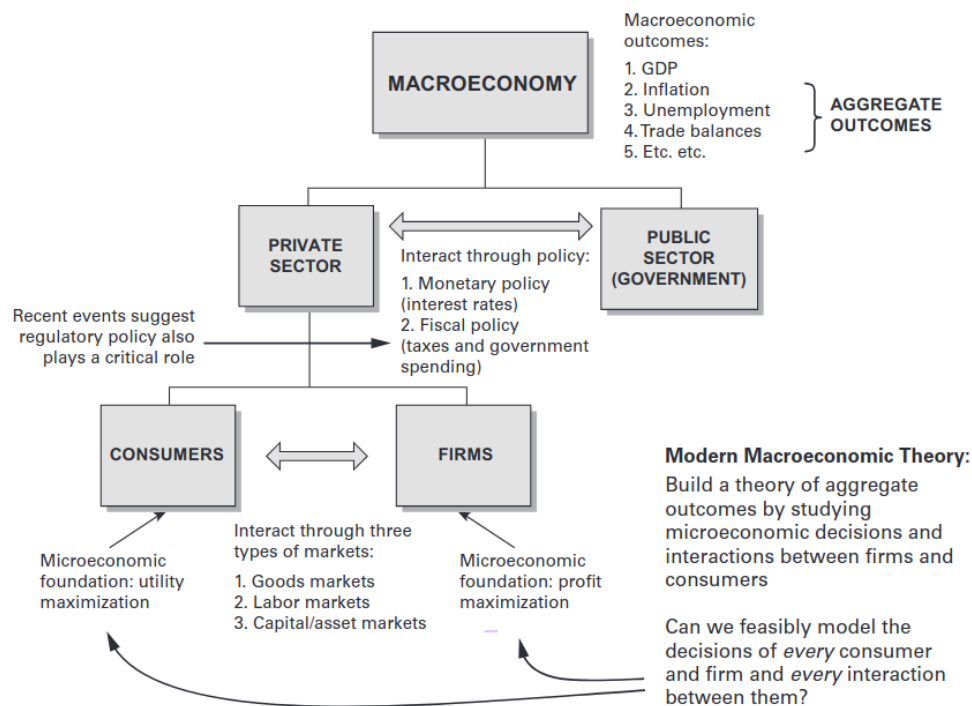


Figure 10.6
Schematic of the overall economy

1.5.1 Supply-Side Economics

1.5.2 The Phillips Curve (PC)

Nominal Wage Rigidity and the Short-Run PC:

With a perfectly competitive labor market, in equilibrium, cyclical unemployment is zero (unemployment rate is equal to the natural rate of unemployment). Suppose there is a negative demand shock that shifts the aggregate demand to the left (price level drops), labor demand then shifts to the left. With a perfectly competitive labor market, wage rate decreases from w^* to w^{new} to clear the market (cyclical unemployment is still zero). Thus it would seem that it is impossible for us to generate a negative relationship between inflation and unemployment wherein cyclical unemployment never changes.

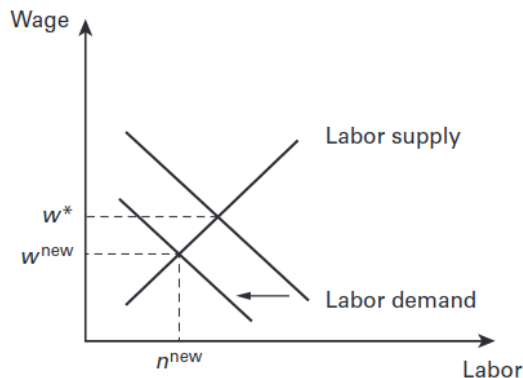


Figure 12.4

After a protracted length of time during which the wage remains stuck above the market-clearing wage, it is likely that firms will win wage reductions (or, at the very least, a threat of layoffs) in the next round of wage negotiations. A decrease in the wage from w^* to w^{new} thus lowers cyclical employment back to zero and hence the unemployment rate down to the natural rate.

However, there is ample reason to believe that the labor market is far from perfectly competitive. The strongest evidence is that wages tend to move very slowly over time, even when other macroeconomic events would tend to suggest sharp movement in wages.

Now suppose that there are some features of the labor market that prevent the wage from adjusting immediately to match demand with supply. Given a negative demand shock (price decreases), labor demand shifts to the left but the wage rate remains stuck at w^* (known as wage rigidity or sticky wage). At the wage rate w^* , quantity demanded for labor is less than quantity supplied of labor, and it brings cyclical unemployment to positive. This negative relationship between the inflation rate and the unemployment rate is captured by the (short-run) Phillips Curve.

Long-run PC:

Supposing that contracts that specify wages in advance are the primary source of wage rigidity, then if aggregate demand continues to slump for a protracted period, we would expect the new round of labor contracts negotiated when the original labor contracts expire to feature lower wages. Continuing with our example from figure 12.4, new rounds of wage negotiation yield the lower nominal wage w^{new} , and it brings the cyclical unemployment to zero, i.e., even though inflation stays low, unemployment rate will eventually

gravitate back to its natural rate in the long run. And this leads to a vertical long-run Phillips Curve.

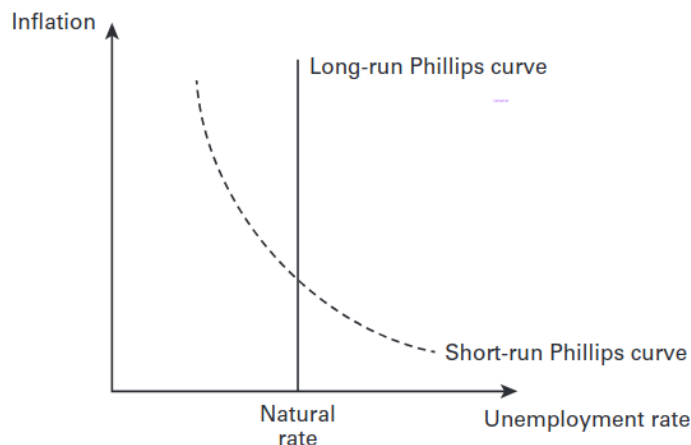


Figure 12.5
Wages adjust in the long run, so there is likely to be no trade-off between inflation and unemployment. This idea is embodied in the vertical long-run Phillips curve.

1.5.3 Real Business Cycle Theory

The two current leading views of business cycles are real business cycle (RBC) theory and New Keynesian economics. Each of these schools of thought has a rich history marked by frequent vigorous debate between them.

RBC theory views periodic expansions and recessions as natural, indeed efficient, responses of the economy to the ups and downs of the state of “technology” of the economy.¹ As such, recessions are not dire events for the economy, but rather natural slowdowns that are preceded by an expansion and that will again be followed by future expansions. In terms familiar from microeconomics, pure RBC theory maintains that the aggregate economy operates perfectly competitively on both the demand side and the supply side.² An important implication of RBC theory therefore is that the government has no role to play in the macroeconomy—that is, neither fiscal policy nor monetary policy can be used to improve the macroeconomic condition.

New Keynesian economics adopts the point of view that there are fundamental market failures in the aggregate economy that render business cycle fluctuations inefficient, specifically periods of lower than potential GDP.³ The important implication of this point of view is that the government may indeed have a role to play in improving macroeconomic conditions.

For the most part, RBC theory has held much less sway among policymakers than has New Keynesian theory. Among theoretical macroeconomists, however, RBC theory is very well-known and well-understood and even provides the foundations for some of New Keynesian theory.¹

Although there are a number of ways in which RBC theory differs from New Keynesian theory, we will focus

on two differences.

- The most important difference by far is that RBC theory eschews (avoid) the idea of sticky prices, while New Keynesian theory embraces it. RBC theory views prices as fully flexible. Even more precisely, RBC theory supposes that perfect competition in all markets is a good starting point for analyzing the macroeconomy.
- Second, RBC theory does not view exogenous shifts of consumption demand as a good description of data but rather “shifts in supply” as the predominant reason for macroeconomic fluctuations.

Measure of Supply Shock Consider the aggregate production function, $A \cdot f(k, n)$, where A capture the technology improvement, called the Total Factor Productivity (TFP). This parameter is usually identified with the Solow residual. This way of measuring technology has the virtue that it does not require taking a stand on what constitute “technology”—that is, it does not require identifying the state of technology of an economy with, say, the number of computers it uses or with the number of PhDs it employs, etc.

The most commonly-used production function in RBC theory is the Cobb-Douglass production function

$$f(k, n) = k^\alpha n^{1-\alpha},$$

in which the parameter α measures the percentage of total GDP that goes toward paying for the costs of capital used in production. More common terminology is that α is capital’s share of output and $1 - \alpha$ is labor’s share of output. Empirical evidence shows that for the US α is about 0.33.

The total output (GDP) is determined by $y = A \cdot f(k, n) = A \cdot k^\alpha n^{1-\alpha}$. We observe y, k, n from the data, and assuming $\alpha = \frac{1}{3}$, so we can calculate A .

The unexplained variation in the technology parameter A should be viewed as **supply shocks**. In the influential study that effectively launched the RBC school of thought, Kydland and Prescott (1982) found that fluctuations in the Solow residual accounted for well over half of fluctuations in GDP, leading them to conclude that a theory of business cycle could be build with technology as its centerpiece.

How Does the Technology Shock Work? Assumptions

- Perfect competition
- HHs owns capital and labor
- Labor tax rate is always zero

Now suppose there is a temporary increase in A . Then, MPK and MPL increase, i.e., for a profit-maximizing

firm, demand for capital and labor increases (shifts outward) as below. Real wage and rental price of capital for that firm is constant (horizontal line) because a single firm is a price-taker in perfect competitive market.

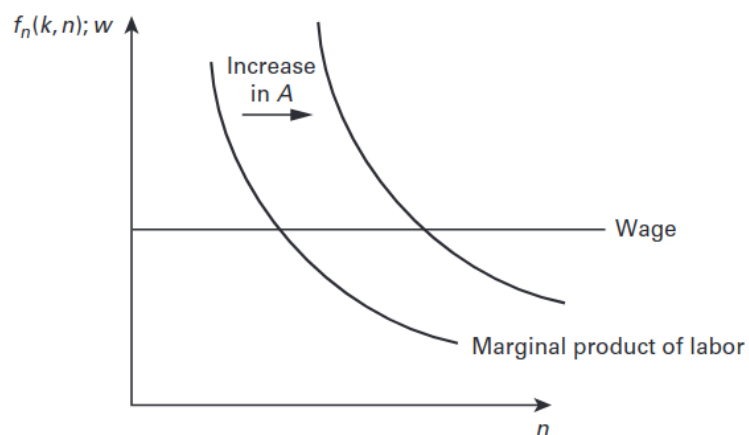


Figure 14.2
Marginal product of labor rises when A rises. For a given wage, the optimal quantity of labor demanded by a firm rises.

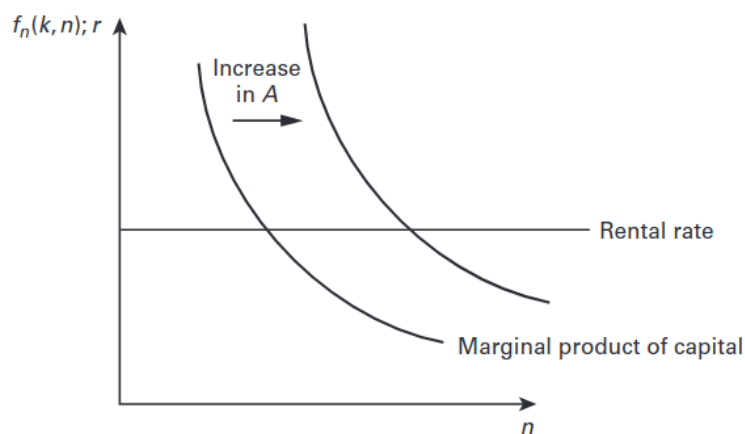


Figure 14.4
Marginal product of capital rises when A rises. For a given rental rate, the optimal quantity of (future) capital demanded by a firm rises.

Now we look at the aggregate labor market and capital market. The right-shift in demand pushes up the price of labor and capital and quantity. The labor demand is derived from the consumption-leisure model. Note that here we assume that the representative consumer is in the upward-sloping portion of the labor supply curve (backward-bended curve).

A rise in A rises the profit-maximizing quantity of (future) capital any given individual firm desires. Thus **the temporary increase in A led to an increase in both future k and current n through its effects on factor prices. And therefore, total output y unambiguously increases in both the current period as well as the future.**

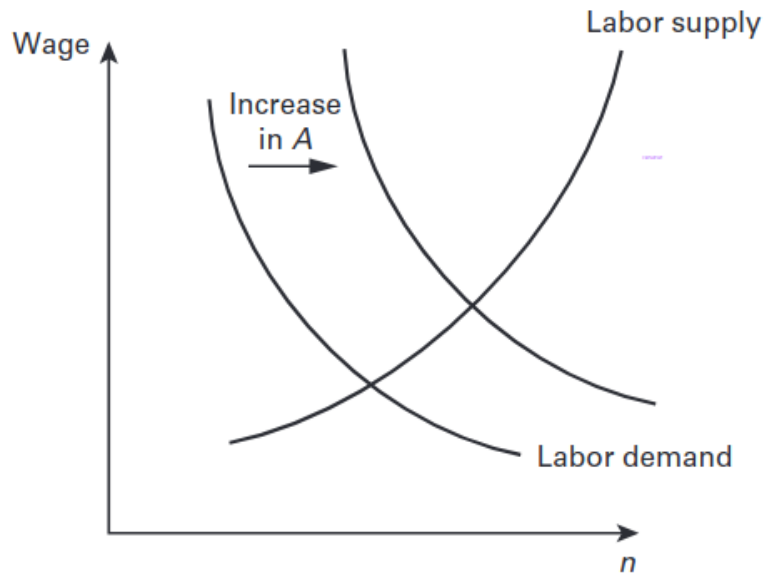


Figure 14.3
Equilibrium real wage rises in the aggregate labor market when technology improves

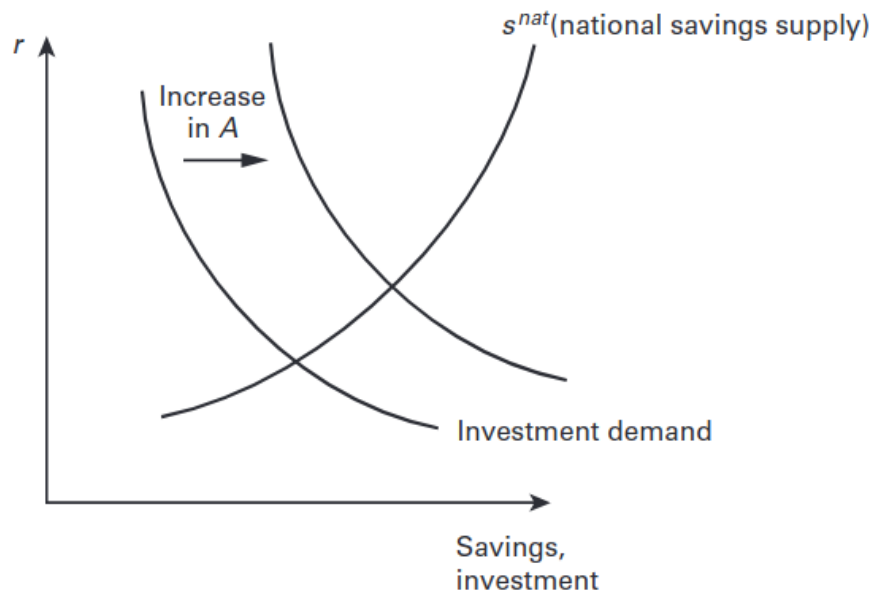


Figure 14.5
Equilibrium rental rate rises in the aggregate funds market when the technology improves.

Now that we have traced out the aggregate effects, we next examine the representative consumer's response along both the static consumption-leisure margin and the consumption-savings margin.

For consumption-leisure margin, as real wage rises, the budget constraint steepens as a result, the new optimal choice of (c, l) in the current period features higher consumption and less leisure (work longer).

For consumption-savings margin, an increase in current labor and future capital results in a higher income in both current and future income, i.e., both y_1 and y_2 increases in the two-period model. Also, as real interest rate rise, the LBC becomes steeper.

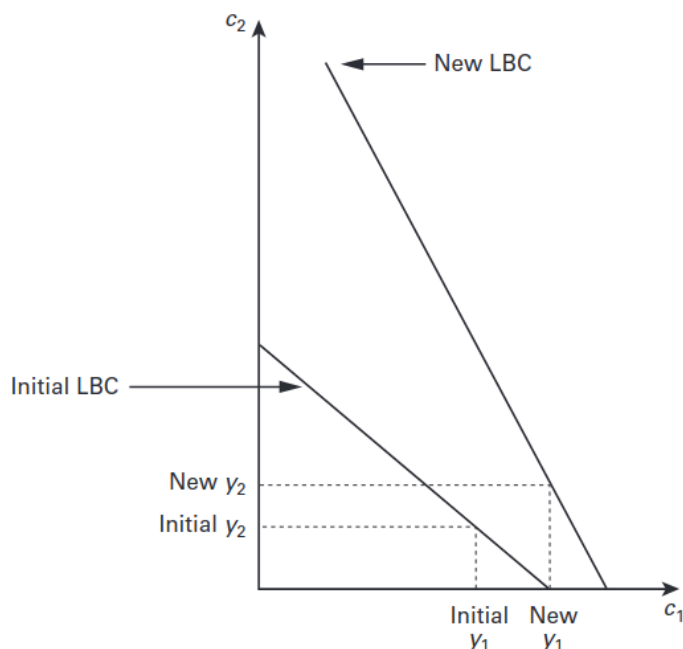


Figure 14.8

Following the rise in A and the resulting rise in r , the consumer's income rises in both period 1 and period 2. The new LBC is steeper than the initial LBC and passes through the new income point.

As shown below, consumption in period 1 increases but does not rise by as much as income rises. The consumer thus optimally spends only part of the gain in period-1 income and save the rest for period 2. This illustrates the important principle of consumption-smoothing. Thus output rises in both period 1 and 2—in the parlance of the RBC literature, the temporary (i.e., for only one period) technology shock leads to a persistent (i.e., for more than one-period) change in output and consumption.

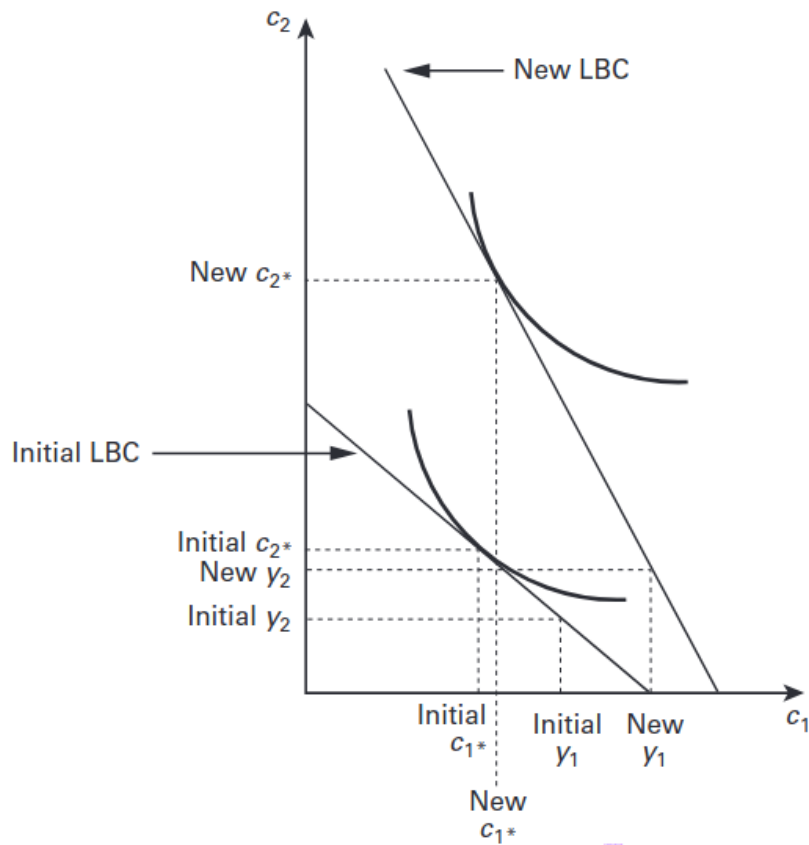


Figure 14.9

Optimal consumption in both period 1 and period 2 rises following the temporary rise in A

1.5.4 New Keynesian Theory

As we will see, microeconomic analysis is at the heart of New Keynesian theory. Unlike most of our discussion of representative-agent macroeconomics, however, the focus of New Keynesian theory is not on the microeconomics of consumer behavior but rather on the microeconomics of firm behavior.

Summary of Assumptions:

- Instead of having just one consumption goods, now consumption is a function of a vast number of goods and services that are substitutable.
- Each of the N differentiated goods is assumed to be produced by a distinct monopolistically competitive firm.

Differentiated Goods and Consumption Aggregator:

In consumption-leisure and consumption-savings model, we assume there is only one object (i.e., “all stuff”) that the consumer purchased in order to obtain utility. The use of a single good is obviously a theoretical simplification. In reality, these goods are somewhat substitutable for each other, e.g., playing badminton vs. playing soccer.

If we believe that there are available a great many options for consumption, each of which is at least a little different from every other option, then one way to reconcile our previous use of a single consumption good with this fact is to suppose that “all stuff” is composed of these great many differentiated goods. And this “all stuff” consumption function can be written as

$$c = c(c^1, c^2, \dots, c^N),$$

where c^i denotes the type- i (or the i th good), while we assume there are N different goods. This function is called the consumption aggregator function.

In theoretical New Keynesian models, the function $c(\cdot)$ is usually assumed to satisfy the following two properties:

$$\begin{aligned}\frac{\partial c(\cdot)}{\partial c^i} &> 0 \\ \frac{\partial^2 c(\cdot)}{\partial (c^i)^2} &< 0.\end{aligned}$$

These two conditions state that total consumption c is an increasing function of consumption of type i and that total consumption c increases at an ever-decreasing rate as consumption of type i increases. For example, the longer time you play badminton, the more total consumption you enjoy—but the more and more you play, the less and less extra total consumption you gain. One example of the consumption aggregator function is

$$c(c^1, c^2, \dots, c^N) = \left[(c^1)^{\frac{1}{\varepsilon}} + (c^2)^{\frac{1}{\varepsilon}} + \dots + (c^N)^{\frac{1}{\varepsilon}} \right]^{\varepsilon},$$

where $\varepsilon \geq 1$. The value ε governs how substitutable, from the point of view of the consumer, the different goods are for each other. With $\varepsilon = 1$, each consumption good is just as good as any other, i.e., they are perfect substitutes for each other. With $\varepsilon > 1$, however, goods are only imperfect substitutes for each other, meaning that they are differentiated to a degree depending on the exact value of ε . A basis for all of New Keynesian economics is the assumption that $\varepsilon > 1$.

Monopolistically Competitive Firms:

Each of the N differentiated goods is assumed to be produced by a distinct monopolistically competitive firm. Recall that a fundamental feature of monopolistic competition is that goods are similar to each other but not completely identical. Also, a firm that produces a differentiated good possesses market power. And this market power manifests itself in the fact that the firm faces a downward-sloping demand curve (instead of a horizontal) and hence a marginal revenue curve that lies strictly below its demand curve. This in turn implies that the firm's profit-maximizing choice ($MC = MR$ determines the quantity to produce, then together with the demand curve to determine the price) features price greater than marginal cost.

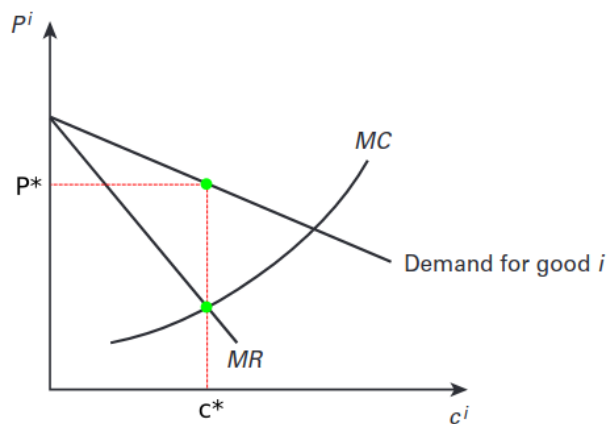


Figure 13.1

A monopolistically competitive firm faces a downward-sloping demand curve for its particular product i . The marginal revenue curve thus lies strictly below the demand curve, and the firm's profit-maximizing choice of output occurs where $MR = MC$. At this optimal quantity, price exceeds marginal cost.

In an extreme case, $\varepsilon = 1$, firms do not have any market power (and thus face perfectly elastic demand curve as the case in the perfect competition, i.e., a horizontal line). Thus setting $\varepsilon = 1$ is one way of shutting off the New Keynesian elements of a New Keynesian model.

The Aggregate Price Level:

In the New Keynesian model, the aggregate price level P is a function of the price for each good,

$$P = P(P^1, P^2, \dots, P^N),$$

we call this function as price-level aggregator. The important feature of this function is that P is strictly

increasing in each of its arguments,

$$\frac{\partial P(\cdot)}{\partial P^1} > 0$$
$$\frac{\partial P(\cdot)}{\partial P^2} > 0.$$

To draw an analogy with the consumer price index (CPI), if the price of any of the goods in the CPI basket rises, the aggregate price level rises.

Aggregate Consumption Demand:

The aggregate consumption demand is not the horizontal summation of each individual demand for good type i , since they are different goods, e.g., we cannot sum the demand for apple and the demand for computer. The consumption aggregator shows the relationship between aggregate consumption demand and the demand for each of the differentiated goods. Recall that total consumption is an increasing function of each type of differentiated good. This fact is all we need conclude that there is a positive relationship between total consumption and consumption of each differentiated good.

The way to interpret the relationship depicted in Figure 13.2 is the following: when some event (e.g., preference shocks and government policy shocks) shifts the consumption demand function, the demand functions facing each firm also shift in the same direction. Because the individual firm's demand functions shifts, the associated marginal revenue function shifts as well, implying a new profit-maximizing choice of price and quantity for each individual firm.

You may notice that different from the RBC theory, New Keynesian theory argues that "demand shocks", coupled with the rest of the theoretical apparatus, are the predominant factor that causes macroeconomic fluctuations. In contrast, RBC theory argues that it is "supply shocks" that are the predominant source of macroeconomic fluctuations.

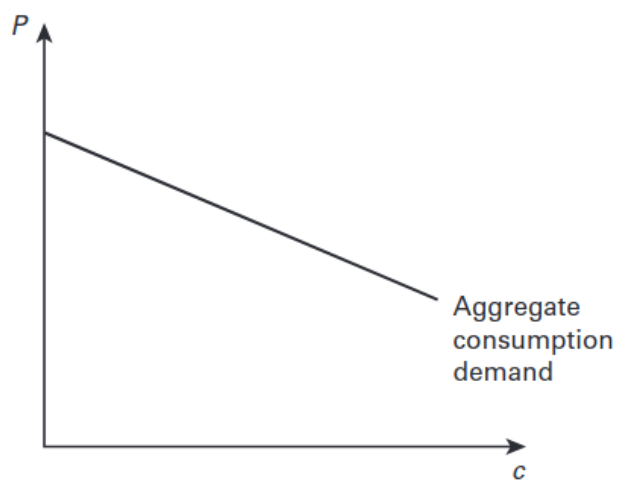
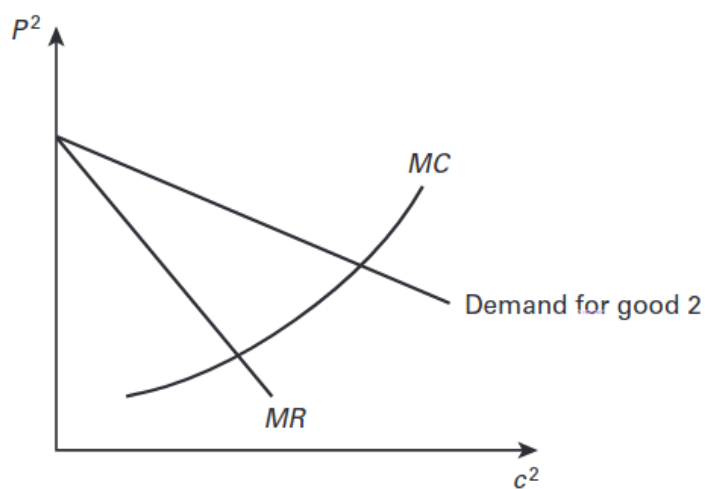
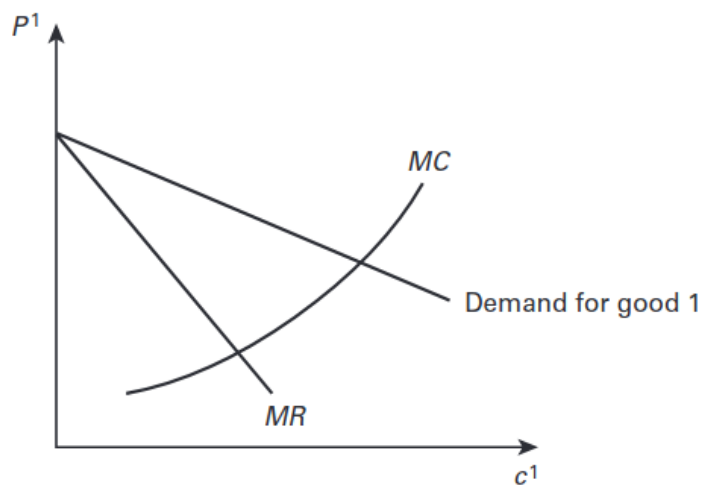


Figure 13.2

Staggered Price-Setting:

Consider the case in which the consumption demand function shifts outward for some reason. This event

in turn causes the demand functions, and hence the associated marginal revenue functions, facing each individual firms to shift outward as well. Assuming the marginal cost functions do not shift, event events in turn cause both quantity c^i and price P^i rise. Because the price of each good rises, the aggregate price level P rises as well.

However, suppose instead that only one of those firms can change its price at the time of the event that shifts the aggregate consumption demand curve. Let's suppose that the rest of the firms have the ability to change price, while firm 1, in contrast, cannot change its price at all.

Figure 13.3 shows the events facing firm 1 following a rise in consumption demand. Firm 1's demand function and MR function both shift outward. If Firm 1 were to change its price, it would choose the price labeled "optimal P^1 if no price stickiness", and the quantity would also rise. However, if firm 1 does not change its price and is forced to continue using the initial optimal P_1 , then its quantity produced will rise by even more (initial price together with the new demand function), as can be read off the new demand function.

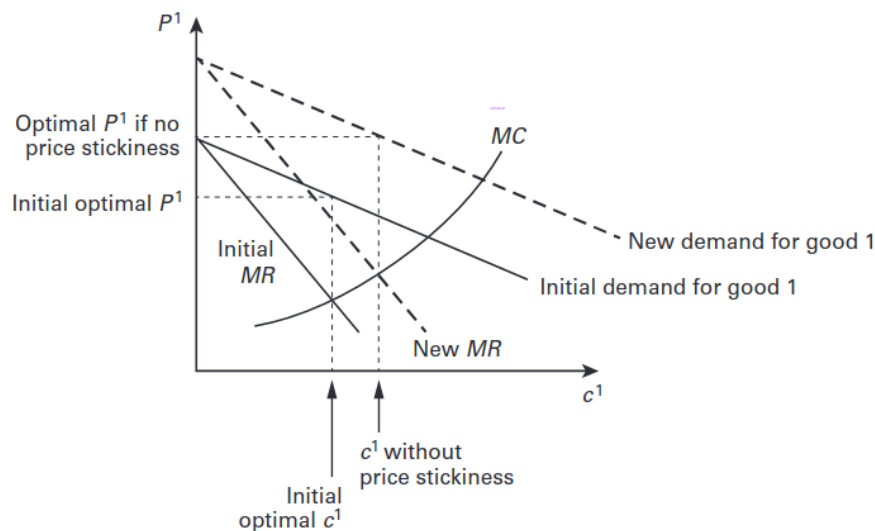


Figure 13.3

With a shift outward of the consumption demand function, the demand function facing firm 1 and hence firm 1's marginal revenue function both shift outward. If firm 1 could change its price, it would raise both its price and quantity produced because the new MR function intersects the MC function at a higher quantity. Associated with such a price increase is a rise in the quantity produced. However, if firm 1 continues with its initial optimal price, then the quantity of good 1 that it produces rises by even more, as can be read off the new demand function.

Implications for Government Policy:

Now we consider how the macroeconomic effects of policy differ depending on whether firm 1 is able to change its price. First, consider the case where all firms can adjust their price. When government policy shifts the consumption demand outward, the aggregate price level P rises because the price of each good rises. Now, suppose that firm 1 does not have the ability to adjust the price while all other firms can do

so. All other firms raise their price in response to a positive demand shock. However, because P^1 does not change, the aggregate price level P does not rise by as much as in the case where firm 1 did change its price. Simultaneously overall consumption c rises by more with firm 1's price held constant because c^1 rises by more in this case.

In the preceding conclusion lie the important implications of New Keynesian theory for government policy. In the presence of staggered price-setting, government policy that raises aggregate consumption demand generates less inflation and a larger increase in production than the same government policy in the absence of staggered price-setting. Thus staggered price-setting gives economic policy makers leverage over real quantities in the economy, more leverage than they would have if all firms could adjust prices simultaneously.

Critique of New Keynesian Theory:

A long-standing criticism of the strand of New Keynesian theory that we have developed is the assumption of staggered price-setting. We have offered no explanation why some firms cannot or does not adjust its price while some others do change its price.

One common justification is the presence of asynchronous menu costs. **Menu costs are the costs incurred by a firm simply by the act of changing prices.** If different sectors of the economy experience menu costs of different magnitudes at different times during the business cycle, then staggered price-setting may rise. While there is little empirical evidence suggesting the magnitudes of these intangible menu cost, it is at least a plausible story.

Strong empirical support for the New Keynesian view comes from data that suggest that government policy, both fiscal and monetary, does have important short-run effects on output. Data generally support the view that government policy is more effective than RBC theory predicts. This last observation alone may be enough justification for studying New Keynesian theory.

2 Key concepts

2.1 Wealth vs. Income

Income is a receipt of money by an individual during some period of time, such as labor income, interest income. An individual's wealth is the level of assets (cash, checking accounts, savings accounts, stock, bonds, etc.) that an individual has in store. It can be negative if an individual is overdrawn on his checking account or otherwise is in debt.

Example::

If you have \$1000 in your savings accounts, then you have \$1000 in wealth. If it generates \$30 interest in period 2, then your interest income is \$30. If you earn a labor income of \$10000 in period 2, your total income is \$10030.

2.2 First-Order Condition

A first-order condition with respect to any particular variable mathematically describes how a maximum is achieved by optimally setting/choosing that particular variable, taking as given the setting/choices for all other variables.

3 Dynamic Consumption-Savings Framework

In this section, we will focus on the study of intertemporal (literally, “across time”) choices of individuals. We first ignore leisure and labor altogether.

Consider a two-period model.

3.1 Assumptions

- Each individual plans economic events for two time periods: present (period 1) and future (period 2). There is no period 3! And every individual knows it! We will relax this assumption later.
- Individual has no control over his or her income. Each individual will receive labor income in each of the two periods.
- Individual has to make a choice about consumption in each of the two periods.
- Saving or borrowings (dissaving) are allowed during period 1.

3.2 Model**3.2.1 Period-by-Period Budget Constraint**

$$\text{Period 1: } P_1 c_1 + A_1 = (1 + i)A_0 + Y_1 \tag{1}$$

$$\text{Period 2: } P_2 c_2 + A_2 = (1 + i)A_1 + Y_2 \tag{2}$$

We must have $A_2 = 0$ as there is no period 3.

Y_1 is labor income received at the beginning of period 1. Each individual begins period 1 with some initial wealth (which may be negative), A_0 . Each individual chooses consumption c_1 in period 1, each unit of which costs P_1 dollars. He also decides how much wealth to carry into period 2, A_1 . A negative A_1 means that the individual is in debt at the beginning of period 2. Nominal interest rate denotes as i .

Private saving in a given time period is being defined as the difference between an individual's total income and his total expenditures, consumption and taxes (we ignore taxes for now). From the above, the total income is the sum of labor income and interest income,

$$TotalIncome_1 = Y_1 + iA_0.$$

Thus saving in period 1 is

$$S^{priv} = iA_0 + Y_1 - P_1c_1$$

If we rearrange equation (??),

$$A_1 - A_0 = iA_0 + Y_1 - P_1c_1,$$

it says saving is the change in wealth during the course of a time period.

3.2.2 Lifetime Budget Constraint

A_1 appears in both equation (??) and (??). Rearrange equation (??) by writting A_1 as a function of all other terms, then put it into equation (??), we get.

$$P_1c_1 + \frac{P_2c_2}{1+i} = Y_1 + \frac{Y_2}{(1+i)} + (1+i)A_0, \quad (3)$$

which is the lifetime budget constraint (LBC). If we write c_2 as a function of c_1 , we get

$$c_2 = -\frac{P_1(1+i)}{P_2}c_1 + \frac{1+i}{P_2}Y_1 + \frac{Y_2}{P_2} \quad (4)$$

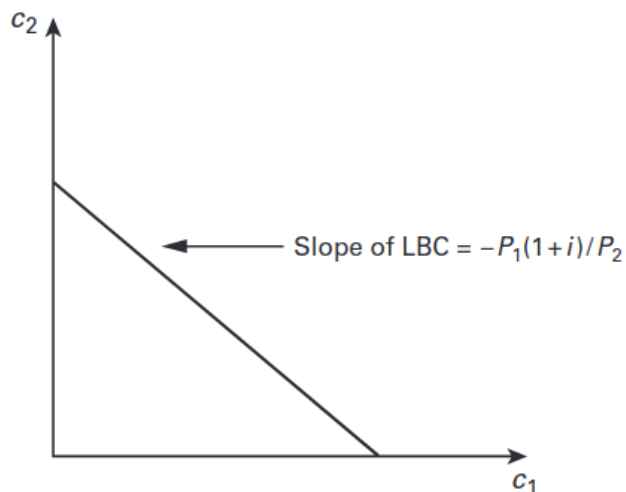


Figure 3.4
Lifetime budget constraint (LBC) of the individual, with the simplifying assumption that $A_0 = 0$

3.2.3 Optimal Intertemporal Choice (Nominal Term)

The optimal choice of consumption is the tangent point between the indifference curve and the LBC. If we assume individuals consume more than they earn in period 1 ($\frac{Y_1}{P_1} < c_1$), then they must consume less than what they would earn in period 2 ($\frac{Y_2}{P_2} > c_2$), so they can pay the debt accumulated in period 1. The figure below illustrate this situation.

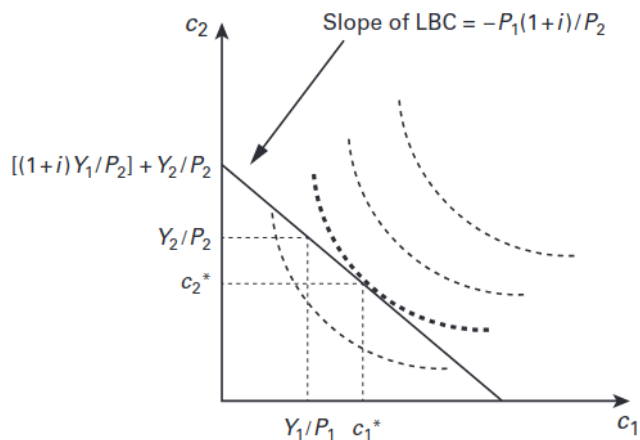


Figure 3.5
Interaction of the individual's LBC and his/her preferences (represented by the indifference map) determine the individual's optimal consumption over time, here c_1^* in period 1 and c_2^* in period 2.

3.2.4 Consumption-Savings Model in Real Units

Recall the LBC in nominal terms,

$$P_1 c_1 + \frac{P_2 c_2}{1+i} = Y_1 + \frac{Y_2}{(1+i)} + (1+i)A_0. \quad (5)$$

To get the real unit,

$$c_1 + \frac{P_2}{P_1} \frac{c_2}{1+i} = \frac{Y_1}{P_1} + \frac{Y_2}{P_1(1+i)} + (1+i) \frac{A_0}{P_1}, \quad \text{divide the nominal LBC by } P_1 \quad (6)$$

$$c_1 + \frac{P_2}{P_1} \frac{c_2}{1+i} = \frac{Y_1}{P_1} + \frac{P_2}{P_1} \frac{Y_2}{P_2(1+i)} + (1+i) \frac{A_0}{P_1}, \quad (7)$$

$$c_1 + c_2 \frac{1+\pi_2}{1+i} = y_1 + y_2 \frac{1+\pi_2}{1+i} + (1+i) \frac{A_0}{P_1}, \quad \text{note, } \frac{P_2}{P_1} = 1 + \pi_2 \quad (8)$$

$$c_1 + c_2 \frac{1+\pi_2}{1+i} = y_1 + y_2 \frac{1+\pi_2}{1+i} + (1+i) \frac{P_0}{P_1} \frac{A_0}{P_0}, \quad (9)$$

$$c_1 + c_2 \frac{1+\pi_2}{1+i} = y_1 + y_2 \frac{1+\pi_2}{1+i} + \frac{1+i}{1+\pi_1} a_0 \quad (10)$$

$$c_1 + \frac{c_2}{1+r} = y_1 + \frac{y_2}{1+r} + a_0(1+r). \quad (11)$$

If we write c_2 as a function of c_1 while assuming $a_0 = 0$ (for simplicity),

$$c_2 = -(1+r)c_1 + (1+r)y_1 + y_2, \quad (12)$$

the slope of the LBC now is $-(1+r)$.

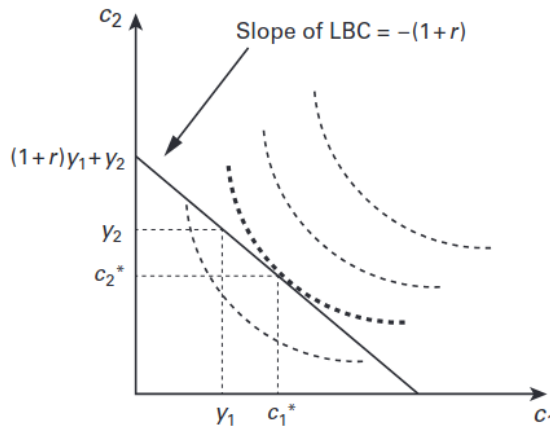


Figure 4.2

Interaction of the individual's LBC (in real terms) and his preferences (represented by the indifference map) determine the individual's optimal consumption over time, c_1^* in period 1 and c_2^* in period 2. The individual begins period 1 with $a_0 = 0$.

Note, **the LBC must pass through (y_1, y_2)** . Now let's consider what happens to this optimal choice if the real interest rate r rises, while the labor income y_1 and y_2 both remain constant. When r rises, the LBC becomes steeper (see below).

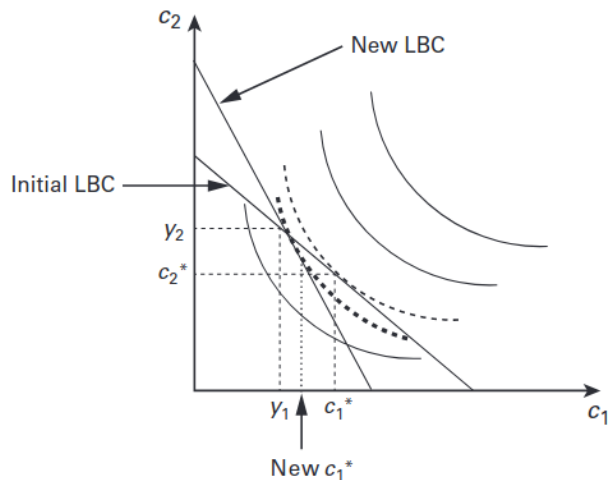


Figure 4.3

If at the initial real interest rate the individual chose to be a debtor at the end of period 1, then a rise in the real interest rate necessarily lowers consumption in period 1, implying that savings during period 1 has increased (or, equivalently, as shown, dissaving has decreased).

Thus, by assume an individual is a debtor in period 1 ($y_1 < c_1$), the models predicts a decrease in consumption, and thus an increase in savings, in period 1 when real interest rate rises.

However, if we assume an individual is a saver in period 1 ($y_1 > c_1$), the model predicts that a rise in r could either increase or decrease consumption. See the two figures below.

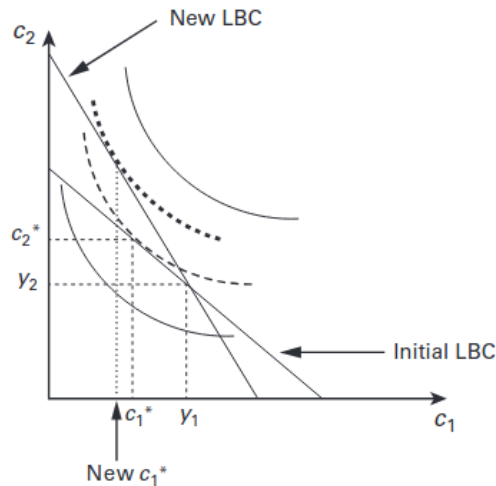


Figure 4.5

If the initial situation is such that the individual optimally chose to be a saver at the end of period 1, then a rise in the real interest rate may cause his savings in period 1 to increase ...

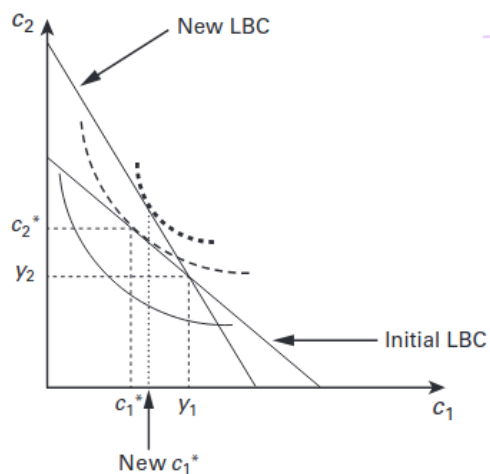


Figure 4.6

... or decrease, depending on the shape of his indifference map (i.e., depending on exactly what functional form his utility function has). Thus, for an individual who optimally initially chooses to be a saver during period 1, it is impossible to determine theoretically in which direction his savings changes if the real interest rate rises.

3.3 Results

Real Interest Rate and Savings:

- If an individual is initially a debtor (consumption is greater than income) at the end of period 1, then a rise in the real interest rate necessarily increases his savings during period 1.
- If an individual is initially a saver at the end of period 1, then a rise in the real interest rate may increase or decrease his savings during period 1.

Yet theory cannot guide use as to how private savings at the macroeconomic level responds to a rise in the real interest rate! Many empirical studies conclude that the real interest rate in fact has a very weak effect, if any effect at all, on private savings behavior.

The studies that do show that real interest rates do influence savings almost always conclude that a rise in the real interest rate leads to a rise in savings.

3.4 Lagrangian and FOC

3.4.1 Lifetime vs. Sequential Lagrange formulation

In analyzing multi-period models using Lagrangians, it turns out we have two alternative and distinctly useful ways of proceeding: an approach we will refer to as a lifetime Lagrange formulation and an approach we will refer to as a sequential Lagrange formulation.

For a simple two-period model, the lifetime Lagrange formulation is essentially nothing more than a formal mathematical statement of the graphical analysis we have already conducted. It emphasizes, as the terminology suggests, that consumers can be viewed as making lifetime choices. The sequential Lagrange formulation emphasizes the unfolding of economic events and choices over time, rather than starting from an explicitly lifetime view. In the end, the sequential approach will bring us to exactly the same conclusion(s) as the lifetime approach; the sequential approach will thus seem like a more circuitous mode of analysis.

Why sequential Lagrange formulation:

We introduce the sequential Lagrangian approach, however, for two reasons. One reason is that when we soon extend things to an infinite-period model, in which graphical analysis becomes quite infeasible, the lifetime Lagrangian formulation (which, as just stated, is really just a mathematical formulation of analysis that can otherwise be carried out purely graphically) inherently becomes a bit less interesting.

A second, and quite related, reason that sequential Lagrangian analysis is of interest is that it will allow us to explicitly track the dynamics of asset prices over time as macroeconomic events unfold over time.

3.4.2 Lifetime Lagrangian Formulation

Assume $A_0 = 0$,

$$u(c_1, c_2) + \lambda \left[Y_1 + \frac{Y_2}{1+i} - P_1 c_1 - \frac{P_2 c_2}{1+i} \right] \quad (13)$$

A first-order condition with respect to any particular variable (think in terms of basic calculus here) mathematically describes how a maximum is achieved by optimally setting/choosing that particular variable, taking as given the settings/choices for all other variables.

In terms of the economics of our model, the consumer is optimally choosing both c_1 and c_2 (in order to maximize utility), which requires computing FOCs of the Lagrangian with respect to both c_1 and c_2 .

$$\begin{aligned}\frac{\partial u}{\partial c_1} - \lambda P_1 &= 0, & \text{FOC wrt to } c_1 \\ \frac{\partial u}{\partial c_2} - \lambda \frac{P_2}{1+i} &= 0, & \text{FOC wrt to } c_2,\end{aligned}$$

From the first equation, $\lambda = \frac{u_{c_1}}{P_1}$, plug it into the second equation, we get

$$\frac{u_{c_2}}{u_{c_1}} = \frac{P_2}{P_1} \frac{1}{1+i} \quad (14)$$

$$\frac{u_{c_2}}{u_{c_1}} = \frac{1 + \pi_2}{1+i} \quad (15)$$

$$\frac{u_{c_2}}{u_{c_1}} = \frac{1}{1+r} \quad (16)$$

It says, when the representative consumer is making his optimal intertemporal choices, he chooses c_1 and c_2 in such a way as to **equate his MRS between period-1 consumption and period-2 consumption to the real interest rate**.

3.4.3 Sequential Lagrangian Formulation

In terms of constraints, in the sequential formulation we will impose all the period-by-period budget constraints, rather than the LBC. A review of basic mathematics (see the mathematical appendix at the end of the book) reminds us that it is straightforward to extend the Lagrangian method to handle optimization problems with multiple constraints. All we need to do, once we have identified the appropriate constraints, is associate distinct Lagrange multipliers with each constraint and then proceed as usual.

By assuming $A_0 = A_2 = 0$, the lagrangian is the following,

$$u(c_1, c_2) + \lambda_1[Y_1 - P_1 c_1 - A_1] + \lambda_2[Y_2 + (1+i)A_1 - P_2 c_2]. \quad (17)$$

Note, λ_1 and λ_2 are distinct multipliers, which has nothing to do with each other.

FOCs in period 1:

Note, in period 1, consumers choose c_1 and A_1 to maximize the utility, and therefore we differentiate the

lagrangian wrt c_1 and A_1 , and get

$$\begin{aligned}\frac{\partial u}{\partial c_1} - \lambda_1 P_1 &= 0, & \text{wrt } c_1 \\ -\lambda_1 + \lambda_2(1+i) &= 0, & \text{wrt } c_2\end{aligned}$$

FOCs in period 2: We are supposed to differentiate wrt c_2 and A_2 , since $A_2 = 0$, we only differentiate the lagrangian wrt c_2 ,

$$\frac{\partial u}{\partial c_2} - \lambda_2 P_2 = 0$$

Combine the three FOCs, we get

$$\frac{u_{c_1}}{u_{c_2}} = \frac{(1+i)P_1}{P_2} \tag{18}$$

$$= 1 + r, \tag{19}$$

which is the same as what we got using the LBC.

3.5 Solving for Numerical Choice

We cannot solve for the numerical value of c_1 and c_2 using only the consumption-savings optimality condition

$$\frac{u_{c_1}}{u_{c_2}} = (1+r), \tag{20}$$

even when we know the form of the utility function. For example, if we assume, for simplicity, the a log utility function,

$$u(c_1, c_2) = \ln c_1 + \ln c_2,$$

equation (20) becomes

$$\frac{c_2}{c_1} = 1 + r.$$

However, we cannot solve for c_1 and c_2 . We need to use both the optimality condition and the budget constraint,

$$\begin{aligned}\frac{u_{c_1}}{u_{c_2}} &= (1+r), \\ P_1 c_1 + \frac{P_2 c_2}{1+i} &= Y_1 + \frac{Y_2}{1+i}.\end{aligned}$$

Example (case1):

Suppose that the lifetime utility function is log utility. And also assume that $P_1 = P_2 = 1$, $A_0 = 0$ and $r = 0.1$.

Further, assume $Y_1 = 2, Y_2 = 11$, then the optimal consumption in period 1 and 2 would be

$$c_1^* = 6, c_2^* = 6.6.$$

Example (case2): Now if we assume this consumer earn more in period 1, $Y_1 = 7, Y_2 = 5.5$, the optimal consumption is still

$$c_1^* = 6, c_2^* = 6.6.$$

Clearly, the lifetime path of optimal consumption is the same, despite the large difference between the case 1 lifetime income path ($Y_1 = 2, Y_2 = 11$) and the case 2 lifetime income path ($Y_1 = 7, Y_2 = 5.5$).

This example demonstrates the two different facets of consumption smoothing.

- Individuals prefer their consumption across time to not vary very much, (6 and 6.6 are close). This result arises due to strictly increasing and strictly concave lifetime utility.
- Despite the two very different income scenarios in the example, optimal c_1 and c_2 are identical. This is due to the ability of the individual to borrow as much as he or she wants during period 1.

If we force the consumer not to borrow in period 1, the case 1 consumption outcomes would be quite different: we would have $c_1 = 2$ and $c_2 = 11$ as the credit-constrained optimal choices.

4 Intertemporal Fiscal Policy (Ricardian vs. Non-Ricardian policy)

Think about this question: whether government spending and taxes affect real interest rates.

The intersection of the upward-sloping savings curve and the downward-sloping investment curve determines the equilibrium real interest rate in the economy. National savings is defined as the sum of private and government savings,

$$S^{national} = S^{private} + S^{government}.$$

4.1 Ricardian Equivalence

Government savings, in many cases, does not typically depend on market real interest rates. Many politically related issues affect government spending and taxation, which in turn directly affect government savings, regardless of what market interest rates might be. Political economy issues are outside the scope of our analysis.

Recall that private savings does depend on the market real interest rate, through its effect on the slope of the consumer's lifetime budget constraint (private savings is an increasing function of the real interest rate). Government spending, though, is much less reliant on market real interest rates because spending and taxation legislation can largely reflect other concerns.

Given a lump-sum tax, the government LBC is

$$g_1 + \frac{g_2}{1+r} = t_1 + \frac{t_2}{1+r}, \quad (21)$$

where g and t refer to government spending and taxes respectively. Consumer LBC is

$$c_1 + \frac{c_2}{1+r} = (y_1 - t_1) + \frac{y_2 - t_2}{1+r}. \quad (22)$$

Combine consumer's and government's LBC,

$$c_1 + \frac{c_2}{1+r} = (y_1 - g_1) + \frac{y_2 - g_2}{1+r} \quad (23)$$

National saving is

$$S_1^{national} = S_1^{private} + S_1^{government} \quad (24)$$

$$= y_1 - t_1 - c_1 + t_1 - g_1 \quad (25)$$

$$= y_1 - c_1 - g_1 \quad (26)$$

From equation (26), the path for government spending (g_1 and g_2) and taxes (t_1 and t_2) must satisfy the government's LBC. Suppose that the government chooses to leave its spending plan unchanged but decides to lower t_1 for some reason, then it must raise taxes in period 2 (t_2) since the government's present value of lifetime spending is unchanged,

$$\Delta t_2 = -(1+r)\Delta t_1 \quad (27)$$

The question we are interested in is **whether this decrease in taxes in period 1 affects national**

savings in period 1. From equation (??), we need to determine how, if at all, consumption c_1 changes due to the change in the timing of taxes.

From equation (??), the only way that the change in the timing of taxes would affect the optimal consumption choice of the individual is if the consumer's LBC is affected. Looking at the RHS of equation (??), suppose t_1 changes by Δt_1 ,

$$\begin{aligned} [y_1 - (t_1 + \Delta t_1)] + \frac{[y_2 - (t_2 + \Delta t_2)]}{1 + r} &= [y_1 - t_1 - \Delta t_1] + \frac{[y_2 - (t_2 - (1 + r)\Delta t_1)]}{1 + r} \\ &= (y_1 - t_1) + \frac{y_2 - t_2}{1 + r} - \Delta t_1 + \frac{(1 + r)\Delta t_1}{1 + r} \\ &= (y_1 - t_1) + \frac{y_2 - t_2}{1 + r}. \end{aligned}$$

The lifetime resources the individual has available for consumption purchases is NOT affected by the change in the timing of lump-sum taxes. Thus, the optimal consumption choice is unchanged, and therefore national savings in period 1 is unaffected by the change of the timing of lump-sum taxes. Thus real interest rate is unchanged. This result is known as Ricardian equivalence.

Ricardian equivalence is the notion that holding fixed a path for government spending, a change in the timing of taxes does not affect the equilibrium real interest rate because it does not affect national savings.

In another word, the decrease in government savings (caused by a tax cut in period 1) is cover by an increase in private savings (In order to pay more taxes in the future, fully rational consumers will simply save the entire tax cut they receive today—which is what it means to say that private savings increases by exactly the amount that government savings decreases), leaving national savings unaffected, which in turn leaves the equilibrium real interest rate unaffected.

Ricardian equivalence thus states that there is no connection between fiscal deficits and real interest rates.

4.2 Distortionary Taxes and The Failure of Ricardian Equivalence

Instead of having a lump-sum tax, now we assume there is a sales tax τ . Then the consumer's budget constraint becomes

$$\begin{aligned} (1 + \tau_1)c_1 + a_1 &= (1 + r)a_0 + y_1 \\ (1 + \tau_2)c_2 + a_2 &= (1 + r)a_1 + y_2, \end{aligned}$$

and the LBC becomes

$$(1 + \tau_1)c_1 + \frac{1 + \tau_2}{1 + r}c_2 = y_1 + \frac{y_2}{1 + r},$$

and rearrange it to

$$c_2 = -\frac{1 + \tau_1}{1 + \tau_2}(1 + r)c_1 + (1 + r)y_1 + y_2.$$

The slope of LBC is now clearly depends on the tax rates τ_1 and τ_2 . Holding fixed a path for government spending, suppose the government decides to lower the taxes rate in period 1. To balance its LBC, the government must raise the tax rate in period 2.

In period 1, when τ_1 decrease, the consumer LBC becomes flatter, and thus consumption in period 1 increases. It decreases the national saving in period 1 (supply shifts to the left), and therefore, real interest rate rises. Ricardian equivalence does not hold.

5 Infinite-Period Framework

5.1 Preferences

The lifetime utility function is defined as the sum of all instantaneous utility function discounted by the discount factor β ,

$$u(c_t) + \beta u(c_{t+1}) + \beta^2 u(c_{t+2}) + \cdots = \sum_{s=0}^{\infty} \beta^s u(c_{t+s}). \quad (28)$$

We use the discount factor β to represent the idea that utility further out in the future is not as valuable as utility closer in time to the present moment. The discount factor β is between zero and one.

- Consumers are perfectly patient if $\beta = 1$
- Consumers are perfectly impatient if $\beta = 0$

The parameter β is meant to be a crude way of modeling the idea of impatience. All else equal, most of us would prefer to have x units of goods right this instant rather than one year from now, and we would probably also prefer to have those x units one year from now rather than two years from now. This is reflected by $\beta < 1, \beta < \beta^2, \beta^2 < \beta^3, \dots$.

5.2 Budget Constraint

Suppose the assets that consumers buy and sell are “shares in the stock market”, and they can receive dividends from holding stocks. Then our infinite-period model’s period- t flow budget constraint is

$$P_t c_t + S_t a_t = S_t a_{t-1} + D_t a_{t-1} + Y_t, \quad (29)$$

where S_t is the nominal price of stock, and D_t is a nominal dividend paid by each share. Asset holding at the end of period t is a_t . Consumption and labor income in period t is c_t and Y_t respectively.

5.3 Lagrangian

The sequential Lagrangian becomes

$$\begin{aligned} & u(c_t) + \beta u(c_{t+1}) + \beta^2 u(c_{t+2}) + \beta^3 u(c_{t+3}) + \dots \\ & + \lambda_t [Y_t + S_t a_{t-1} + D_t a_{t-1} - P_t c_t - S_t a_t] \\ & + \beta \lambda_{t+1} [Y_{t+1} + S_{t+1} a_t + D_{t+1} a_t - P_{t+1} c_{t+1} - S_{t+1} a_{t+1}] \\ & + \beta^2 \lambda_{t+2} [Y_{t+2} + S_{t+2} a_{t+1} + D_{t+2} a_{t+1} - P_{t+2} c_{t+2} - S_{t+2} a_{t+2}] \\ & + \dots, \end{aligned}$$

Note carefully that the $t + 1$ budget constraint in the Lagrangian is discounted by β . This is because everything about period $t + 1$ is discounted when viewing from the perspective of time t , including income and expenditures.

FOC wrt c_t and a_t :

$$\begin{aligned} u'(c_t) - \lambda_t P_t &= 0 \\ -\lambda_t S_t + \beta \lambda_{t+1} (S_{t+1} + D_{t+1}) &= 0. \end{aligned}$$

From the FOC on consumption in period t , $\lambda_t = \frac{u'(c_t)}{P_t}$. From the FOC on consumption in period $t + 1$, $\lambda_{t+1} = \frac{u'(c_{t+1})}{P_{t+1}}$. Insert λ_t and λ_{t+1} into the FOC on shares of stock, we have

$$\frac{u'(c_t) S_t}{P_t} = \beta \frac{u'(c_{t+1}) (S_{t+1} + D_{t+1})}{P_{t+1}}. \quad (30)$$

We can interpret the result in two ways: one is geared toward macroeconomic analysis, the other toward financial market analysis.

5.3.1 Macroeconomic Analysis

First, rewrite equation (30) to highlight the intertemporal marginal rate of substitution,

$$\frac{u'(c_t)}{\beta u'(c_{t+1})} = \frac{S_{t+1} + D_{t+1}}{S_t} \frac{P_t}{P_{t+1}}.$$

The LHS is the intertemporal MRS. The first part of the RHS, $\frac{S_{t+1}+D_{t+1}}{S_t}$, is the holding return of the asset a_t . It measures the gain (or loss) of holding the asset a_t from period t to $t+1$. The second part of the RHS is just $1 + \pi_{t+1}$. And therefore, the RHS can be view as the real rate of return, which works similar to the $1+r$ in the consumption-saving model. The reason $(1+r)$ does not appear explicitly is simply because of the assumption about the available assets we have made here. If we assume there are assets in the environment that pay a nominal interest rate, we will have our $1+r$ on the RHS.

5.3.2 Financial market Analysis

Page 130

5.4 Steady State

We define the steady state of an economy as a situation in which all real variables stop fluctuating over time, i.e., the value of them becomes constant. The steady state of an economy is often referred to as the long-run equilibrium of an economy—think of it, as the “average” or “potential” performance of the economy. To provide ourselves some more notation, in steady state, the level of consumption and real interest rate is the following,

$$c_t = c_{t+1} = c_{t+2} = \cdots = \bar{c}$$

$$r_t = r_{t+1} = r_{t+2} = \cdots = \bar{r}.$$

Consider the expression mentioned above,

$$\frac{u'(c_t)}{\beta u'(c_{t+1})} = 1 + r_t.$$

This expression is nothing more than the infinite-period model’s consumption-savings optimality condition. In a steady state the optimality condition can be expressed as

$$\frac{u'(\bar{c})}{\beta u'(\bar{c})} = 1 + \bar{r}$$

$$\frac{1}{\beta} = 1 + \bar{r},$$

The expression, $\frac{1}{\beta} = 1 + \bar{r}$ is the long-run consumption-savings optimality condition, captures an extremely critical idea embedded in virtually all of modern macroeconomic theory and thus is at the root of a wide range of both academic and policy discussions of macroeconomics.

It says that the real interest rate of the economy is fundamentally tied to the degree of impatience of consumers in the economy. The theoretical upper end of β is $\beta = 1$; if $\beta = 1$, then the long-run real interest rate equals zero. That is, if consumers are perfectly patient ($\beta = 1$), there is no net real return from savings. Suppose instead that $\beta = 0.95$, meaning that consumers are somewhat impatient. The long-run real interest rate in the economy is roughly $\bar{r} = 0.0526$. Now suppose that $\beta = 0.9$, then $\bar{r} = 0.11$.

5.5 Overlapping Two-Period Frameworks

The infinite-period framework is simply an infinite sequence of overlapping two-period frameworks. In period 1 ($t = 1$), consumers make decision on c_1 and c_2 , the slope of the budget constraint is $-(1 + r_1)$. In period 2 ($t = 2$), consumers make decision on c_2 and c_3 , the slope of the budget constraint is $-(1 + r_2)$.

...

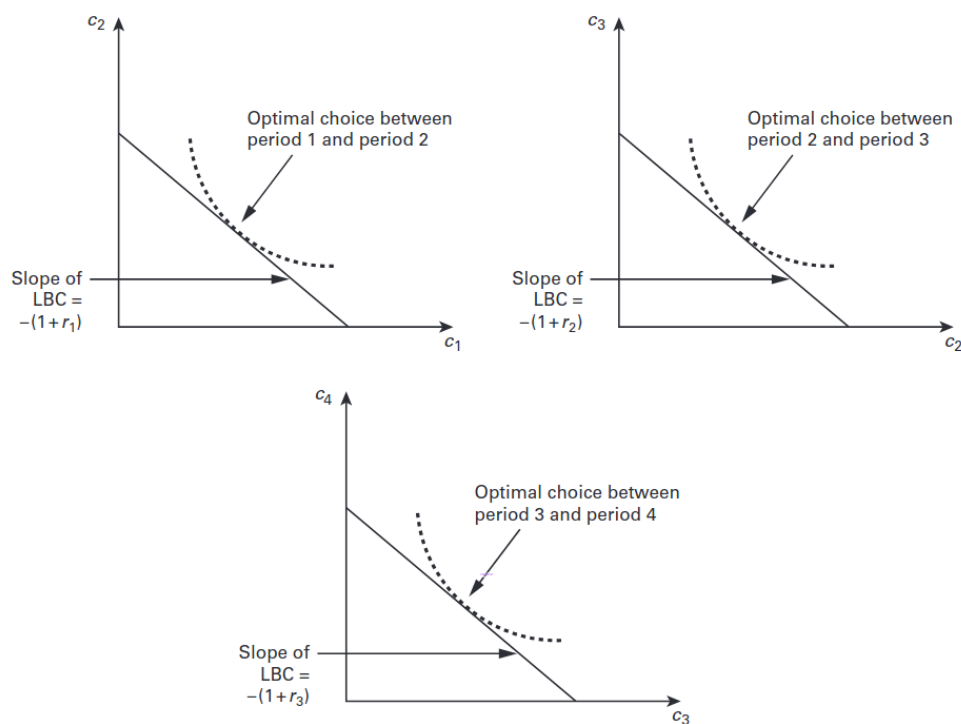


Figure 8.2
Infinite-period framework displayed as a sequence of overlapping two-period frameworks

6 Shocks

The demand and supply relationships in the frameworks we have studied have been developed from microeconomic foundation, which are optimizations of some objective function subject to some constraint function(s).

The demand and supply relationships can shift if either or both the objective function and constraint function(s)

experience some “changes”.

In this section, we study sudden changes in an appropriate utility function if consumer analysis is the focus, and in an appropriate profit function if firm analysis is the focus. These changes in foundations lead to precise shifts in the accompanying demand and/or supply curves.

The changes that we study here are termed shocks. A shock is an unexplained or unexplainable alteration in some basic element of an economic framework, which in turn causes optimal choices to be affected.

6.1 Production Shocks

The introduction of a shock to the production function has the effect that for any given amount of both capital and labor, total output depends on the level of the shock. Such a shock is most commonly interpreted as a “technology shock.”

The most usual way of introducing production function shocks is to suppose that it simply multiplies the production function (parameter A). If we set $A = 1$ always, then we recover the basic two-period firm model we have studied.

If A rises or falls, then the production function scales up or down. This technology shock will be important in our upcoming discussion of real business cycle theory.

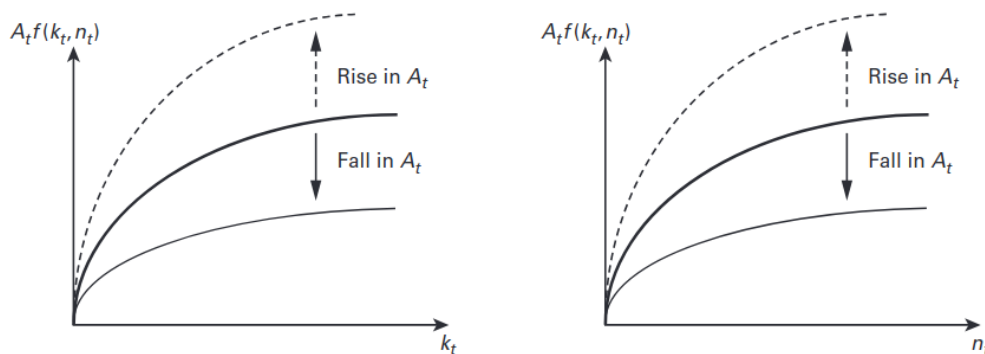


Figure 9.1
An increase (decrease) in A_t causes output to increase (decrease) for any given quantity of capital and labor.

6.2 Preference Shocks

Consider the utility function in consumption-leisure model, $u(c, l)$, where c is consumption and l is leisure. Now introduce a multiplier for consumption, and therefore the utility function becomes $u(Bc, l)$, where B is some given constant over which the individual has no control. Suppose the new utility function takes the

following form,

$$u(Bc, l) = \sqrt{Bc} + \sqrt{l}.$$

Now suppose that all of a sudden B falls from 1 to 0.5 due to some unexplained event. If we assume the optimal consumption given $B = 1$ is $c^* = 10$, then the decrease in B causes consumption part drops from $\sqrt{1 \times 10} \approx 3.16$ to $\sqrt{0.5 \times 10} \approx 2.23$. This means that each unit of consumption is now less valuable in utility terms. And therefore consumers are willing to give up more units of consumption for a given increase in leisure. Thus the indifference curve of the individual would be steeper, as shown below

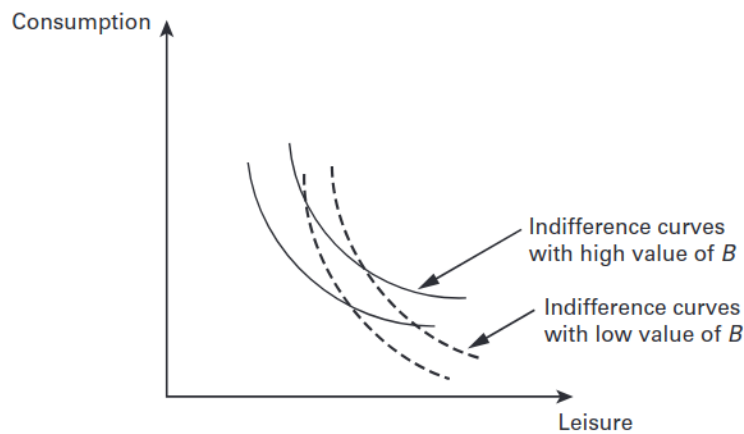


Figure 9.2

As B falls, the individual's indifference map steepens because the individual is willing to trade more consumption for leisure.

Results:

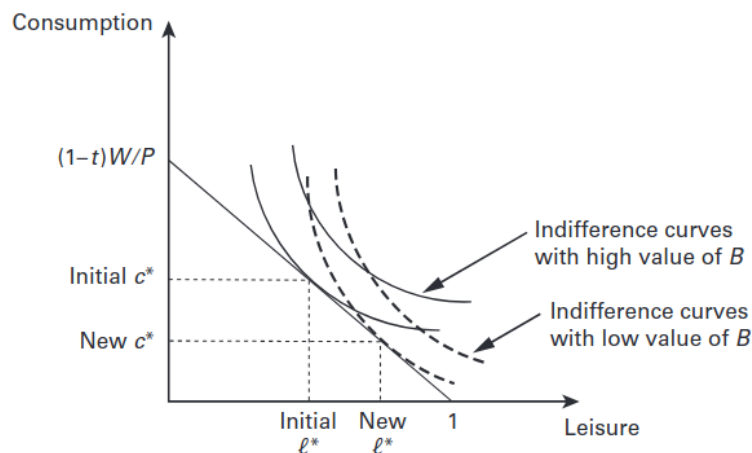


Figure 9.3

Following a decline in B (with the nominal wage W , the nominal price P , and the tax rate t held constant), the new optimal choice features less consumption and more leisure (and hence fewer hours worked).

Given a lower B , what's happening to the optimal consumption and leisure choice?:

With the wage rate W , the price P , and the labor tax rate t all held constant, the new optimal choice features less consumption and more leisure—the latter implying that the individual now works fewer hours

Given a lower B , what’s happening to the labor supply?:

As consumers choose to work fewer hours, the entire labor supply curve must shift inward.

Given a lower B , what’s happening to the consumption demand?:

The change in the optimal choice in figure 9.3 occurs with no change in the price P , i.e., such a reduction in consumption would occur for any given price P . Thus, the consumption demand function shifts inward.

Given a lower B , what’s happening to the saving function?: If we introduce this shock to the consumption-savings model, a decrease in consumption would cause the aggregate savings function to shift.

7 General Equilibrium Macroeconomics

General equilibrium occurs when all markets in the economy are simultaneously in equilibrium. In contrast, partial equilibrium occurs when one market in the economy achieves equilibrium, regardless of whether or not other markets have reached the point at which quantity supplied equates with quantity demanded.

Let’s suppose that there are no distortionary taxes (lump-sum taxes only).

Labor Market:

Let’s take a look at the following pair of optimality conditions:

$$\frac{u_l(c_t, l_t)}{u_c(c_t, l_t)} = w_t, \quad w_t = MPL_t$$

The expression on the left is the consumption-labor optimality condition, which is the labor supply function.

The expression on the right is the firm’s optimality condition regarding hiring, which is the labor demand function.

The equilibrium in the labor market occurs at the intersection of the supply and the demand,

$$\frac{u_l(c_t, l_t)}{u_c(c_t, l_t)} = MPL_t.$$

The equilibrium occurs when the willingness of the consumer side of the economy to trade goods for leisure (which is the MRS between consumption and leisure) exactly equates with the extra (marginal) productivity

of an extra unit of time spent in labor.

Capital Market:

Consider the following pair of optimality condition:

$$\frac{u'(c_t)}{\beta u'(c_{t+1})} = 1 + r_t, \quad r_t = MPK_t.$$

The expression on the left is the consumption-savings optimality condition, which yields the private savings supply function. The expression on the right is the firm's optimality condition regarding investing in new physical capital for the future. Put the demand and supply together,

$$\frac{u'(c_t)}{\beta u'(c_{t+1})} = 1 + MPK_t.$$

Goods Market:

Explain later.

Aggregate supply and aggregate demand.